

CSCI3170

Intro to Database Systems

WK2 Tutorial

Part One: Review

Basic ER-model Notations

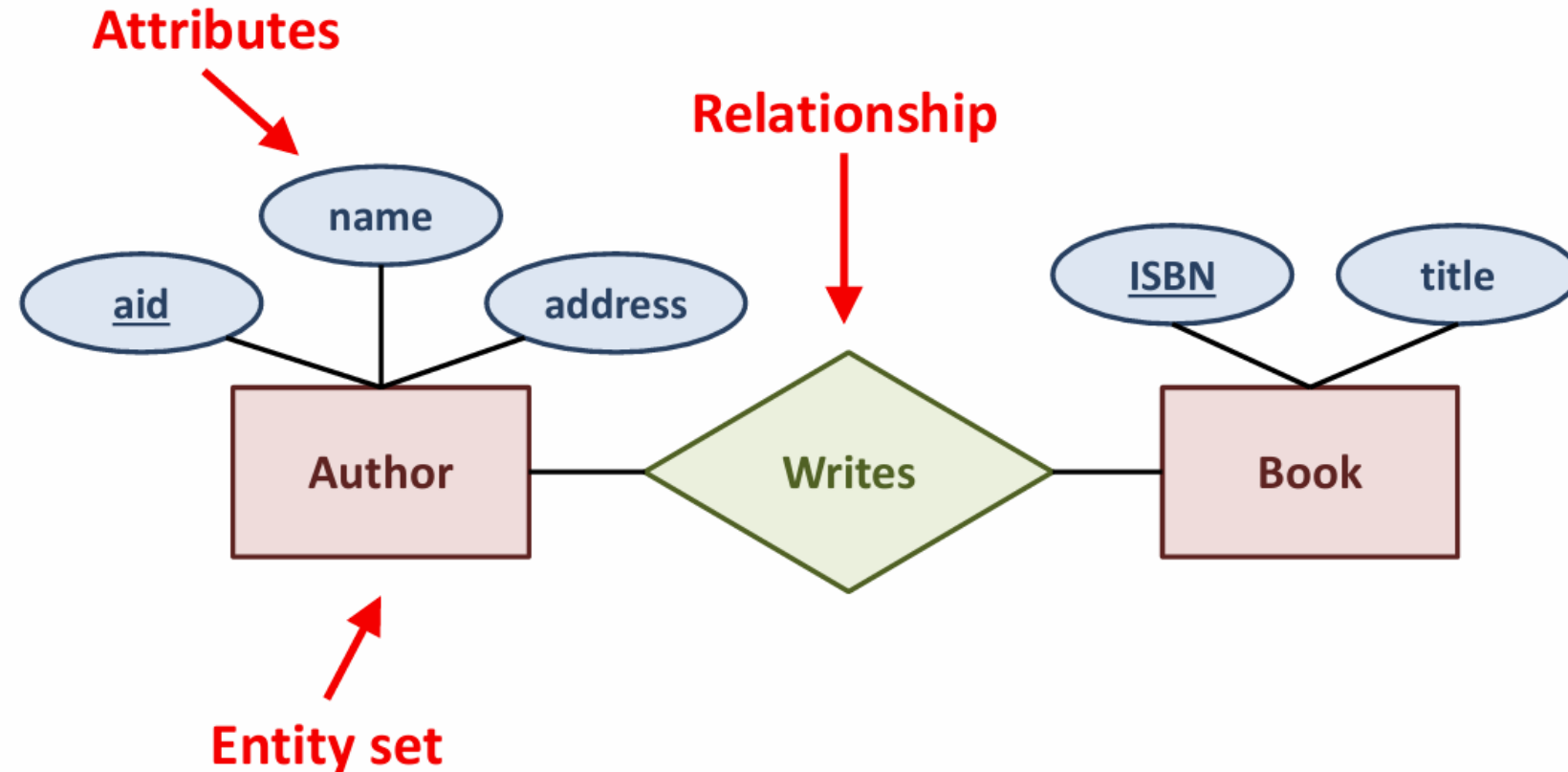
E/R-model

- A conceptual data model
- Real world “objects” are mapped to entities and described by relationships among the entities.

- There are different sets of notations for drawing an ER-diagram
- For our course, you are required to use the notations mentioned in the lecture notes
- **Marks will be deduced if notations other than the one specified in lecture notes are used in assignments, projects and examinations**

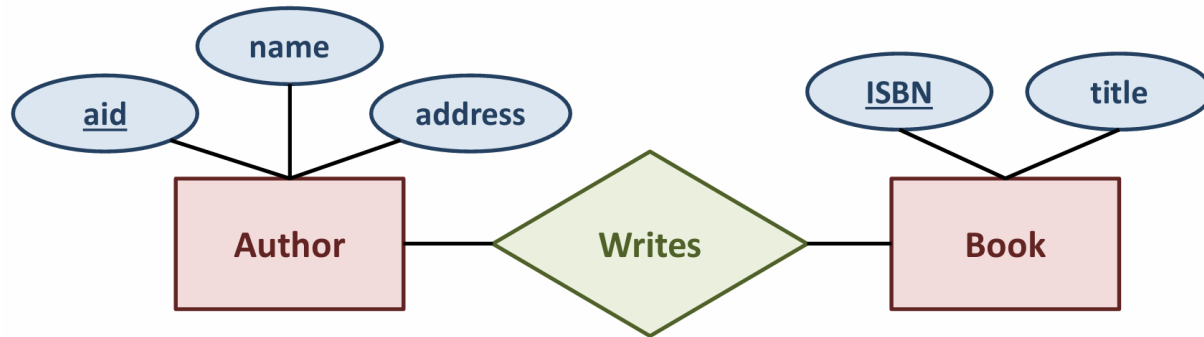
E/R Diagram(s)

- Equivalent graphical representation of an E/R model

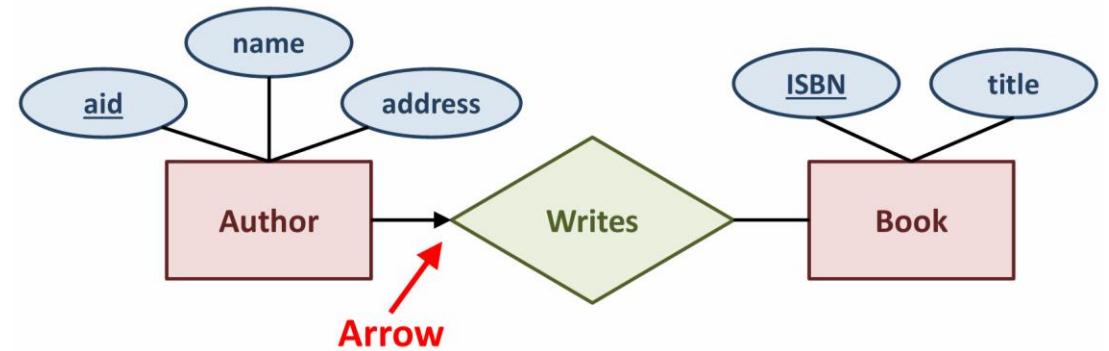


Relationships

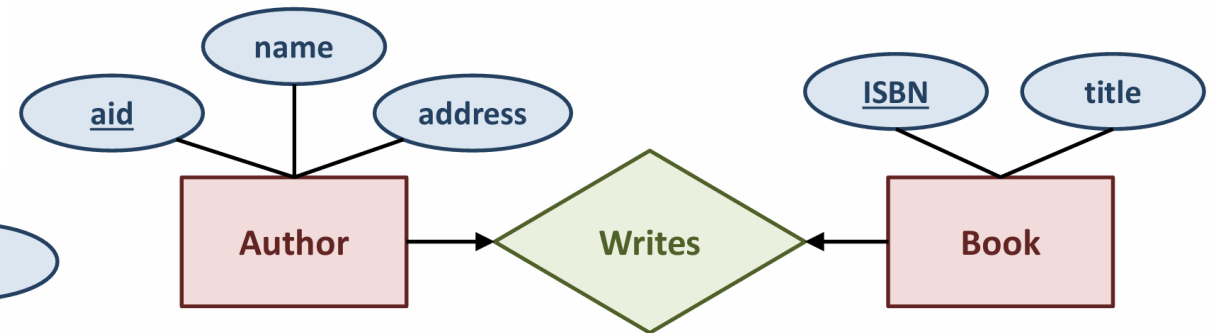
- Arrows indicating an “**at-most-once with other entities**” type restriction on cardinality.
- Default is as many as you want if you want to.



- An author can write **any number of** books
- A book can be written by **any number of** authors



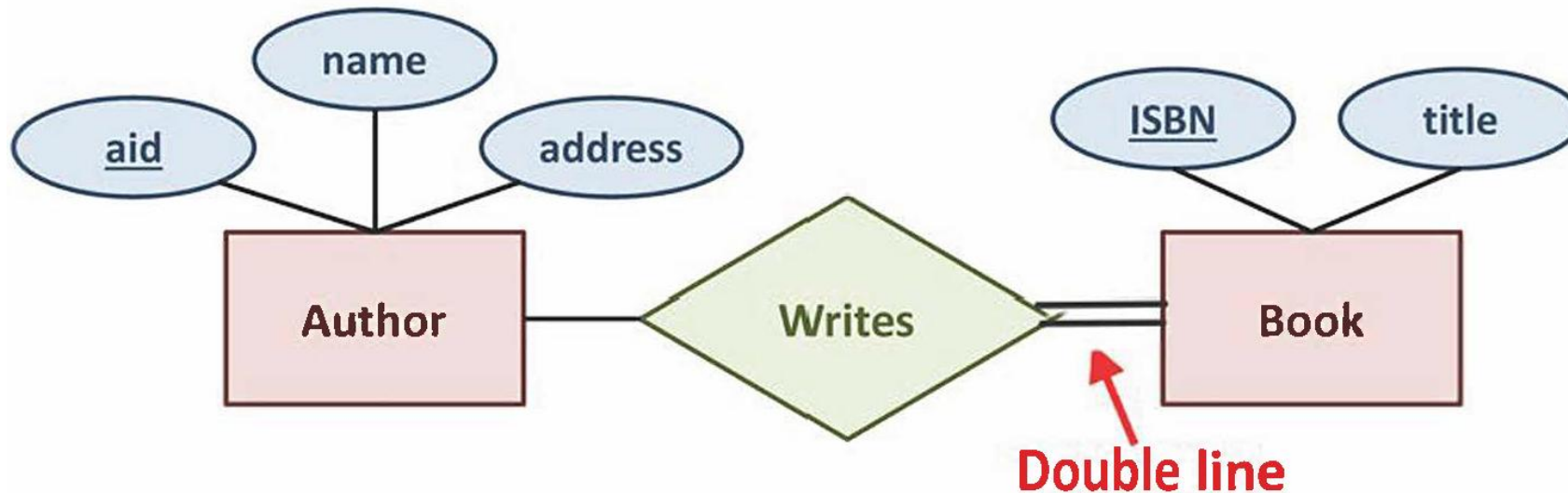
- An author can write **at most one** book
- A book can be written by **any number of** authors



- An author can write **at most one** book
- A book can be written by **at most one** author

Participation (default = partial)

- A double-line indicates “**must participate at least once**” type restriction on participation. (total participation)



- An author can write **any number of** books
- A book must be written by **at least one** author

Demonstrate by Example Library Database

Sample Data Requirements I - Library

- A **book** is uniquely identified by its book-id. For each book, we also record its title, price, and availability.
- A **reader** is uniquely identified by his/her reader-id and we also record his/her name, phone number and address. The address is composed of street and suburb.
- A **publisher** is uniquely identified by its publisher-id. For each publisher, the name is also recorded.
- An **author** is uniquely identified by his/her author-id. For each author, the name, phone number and birth date are also recorded.
- A **reader** can borrow zero or more books and a book can be borrowed by zero or more readers. We need to record starting date and ending date for the borrowing relationship.
- A **publisher** can publish zero or more books and a book is published by exactly one publisher. We also need to record the date of publication.
- An **author** can write zero or more books and a book is written by one or more authors.

[Sample Only] identify entity types

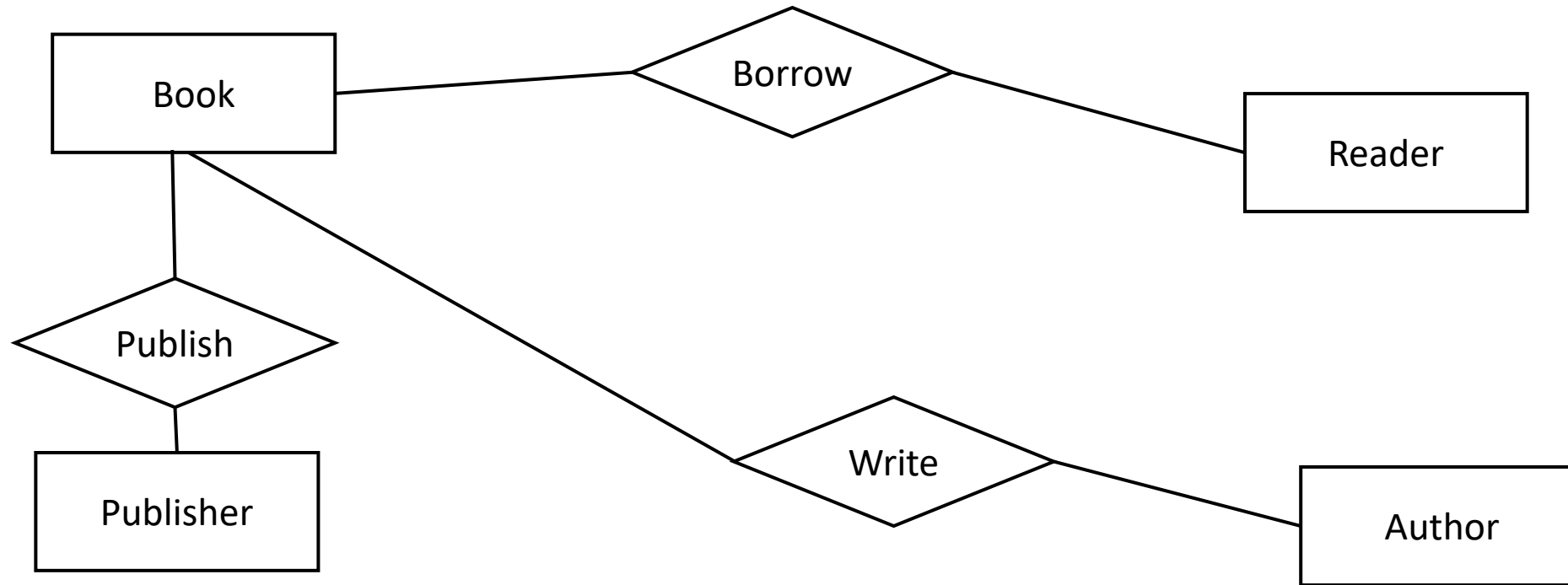
Book

Reader

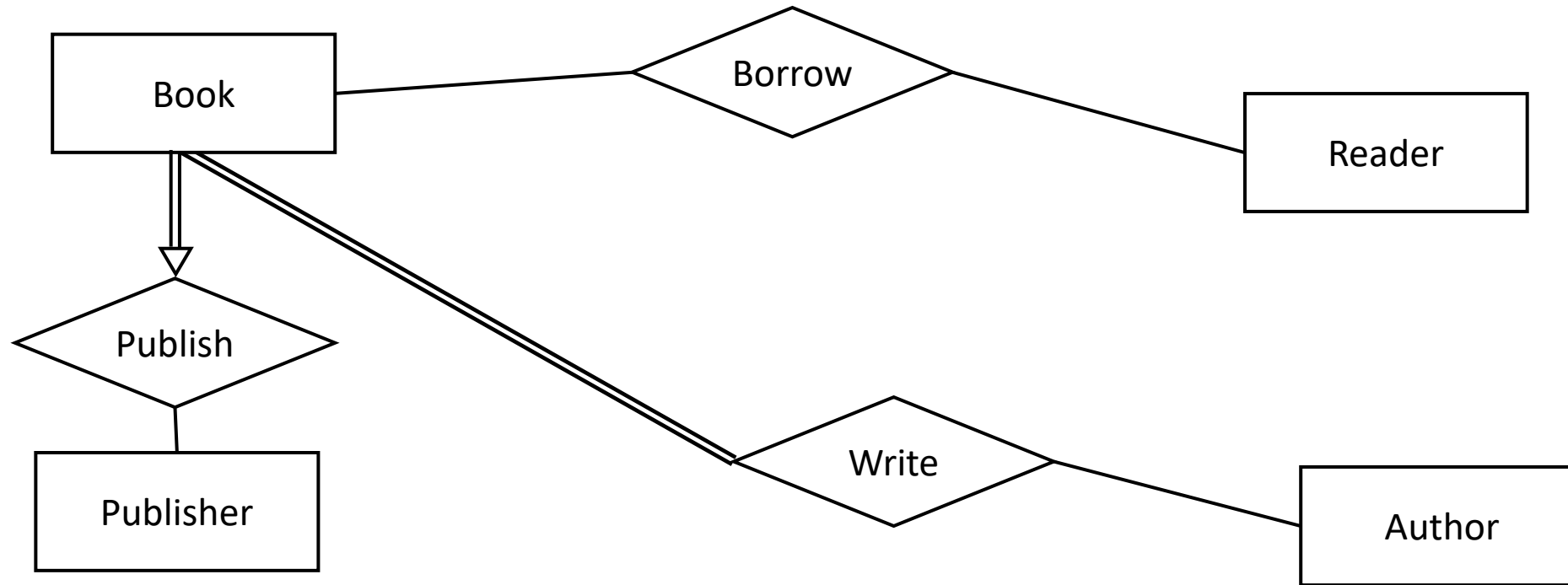
Publisher

Author

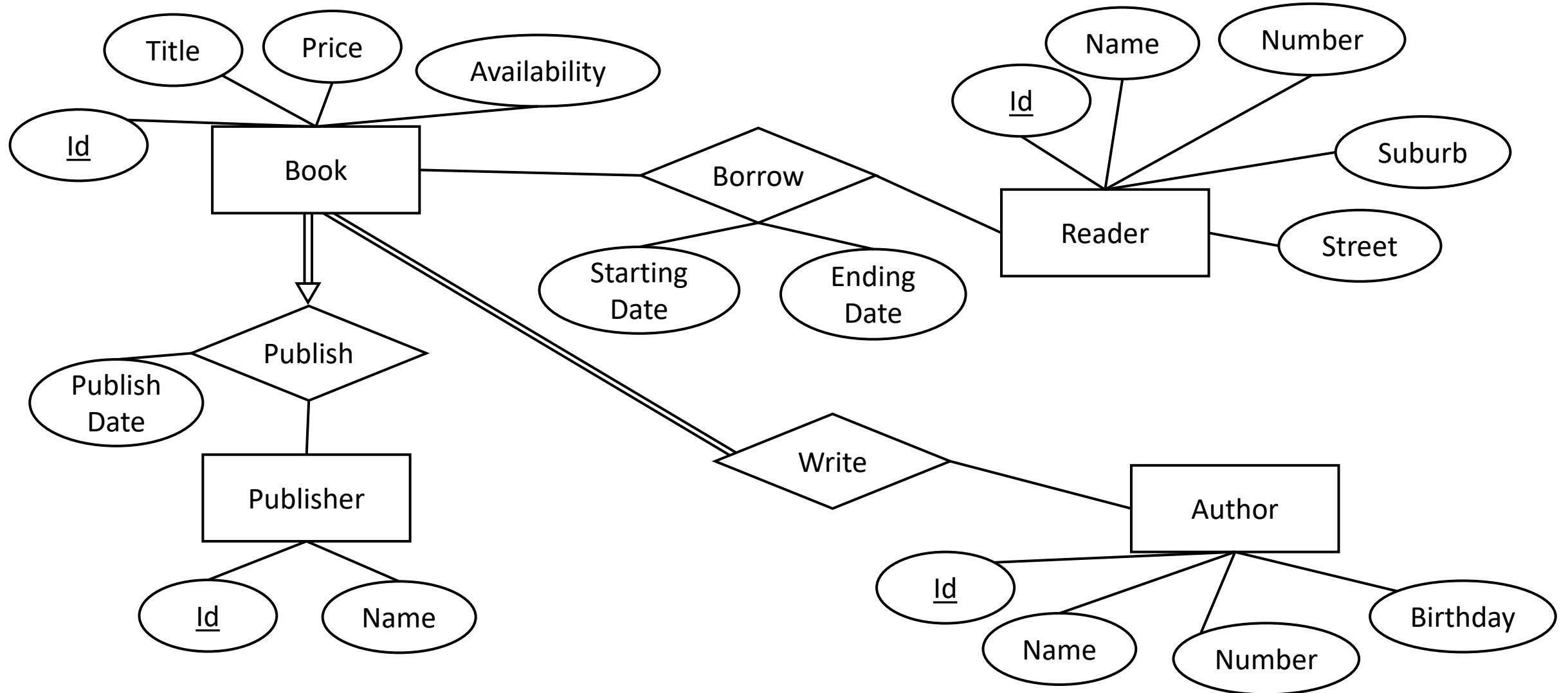
[Sample Only] include relationships



[Sample Only] add rel. constraints



[Sample Only] add all attributes



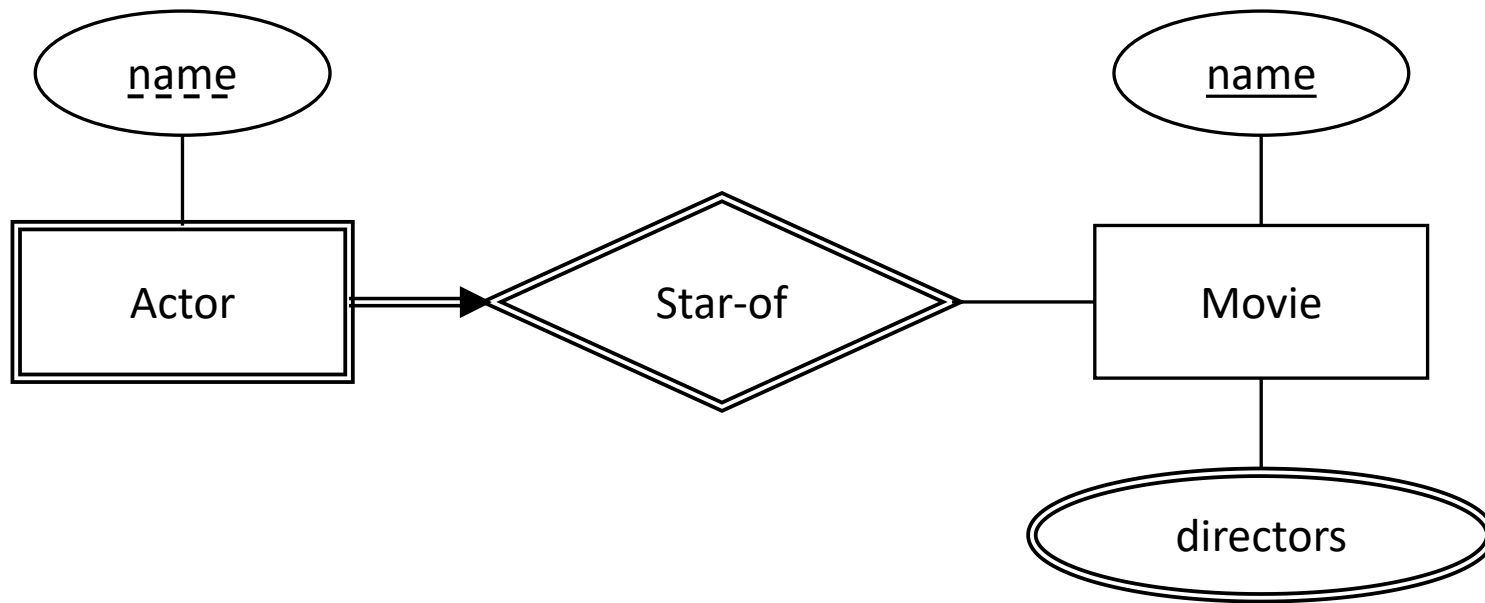
Sample Data Req. II - University

- The university has many departments.
- Each department has a name and a location.
- Multiple instructors work in a department, and an instructor can work in at most one department.
- Each instructor has an ID, a name and an email-address
- For each department, there is a head who is also an instructor.
- An instructor can be the head of at most one department at a time.
- Each department can offer any number of courses, each course is offered by only one department.
- Each instructor can teach any number of courses, a course is taught by one or several instructors.
- Each course has a course-code, and a course-title
- Each student has a student-ID, and a name.
- A student can enrol in any number of courses, and each course can have any number of students.

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Sample E/R Model - Movie Actors

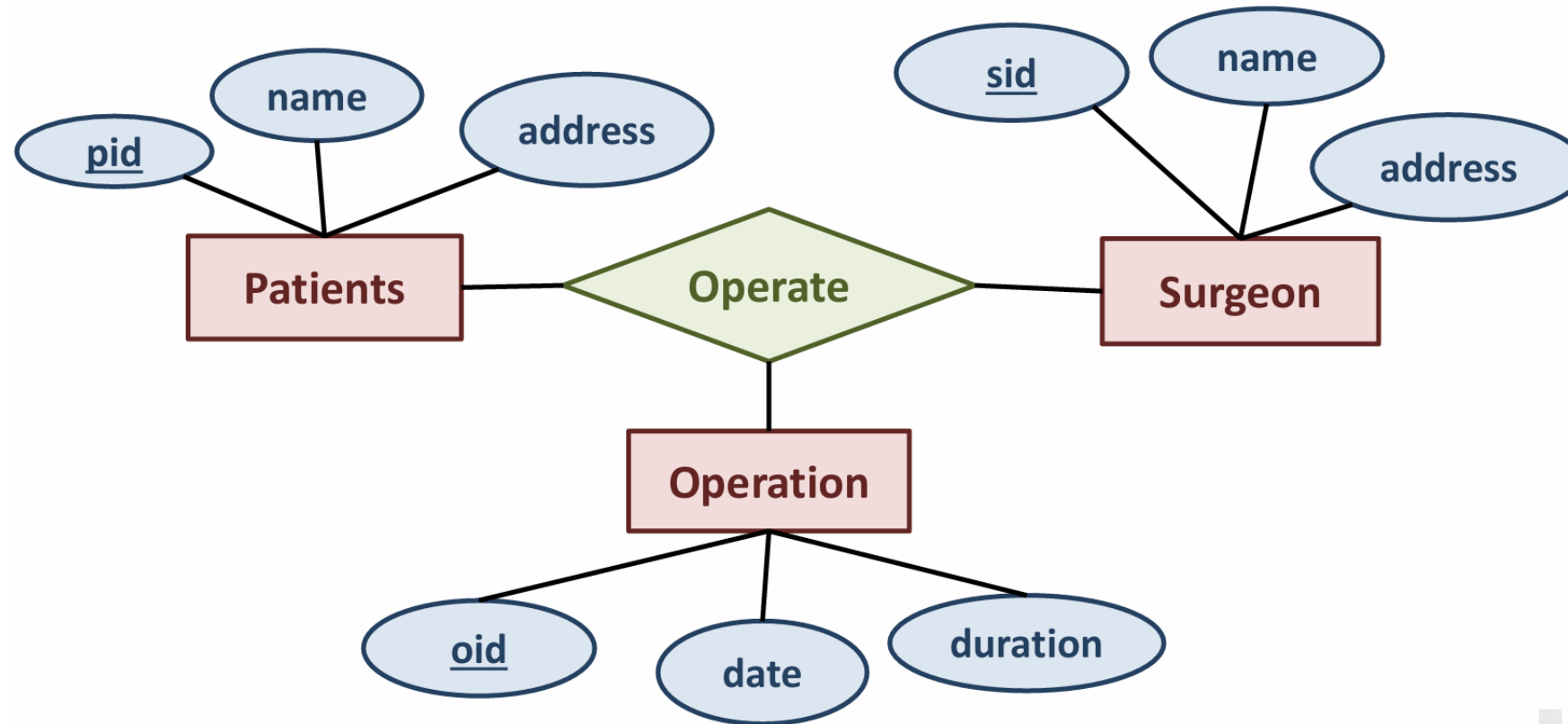
- it is one thing to consider whether the model is realistic; it is another to work with a given modelling of data w.r.t provided constraints.



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Ternary relationships

- Sometimes, three (or more) entity sets are related.



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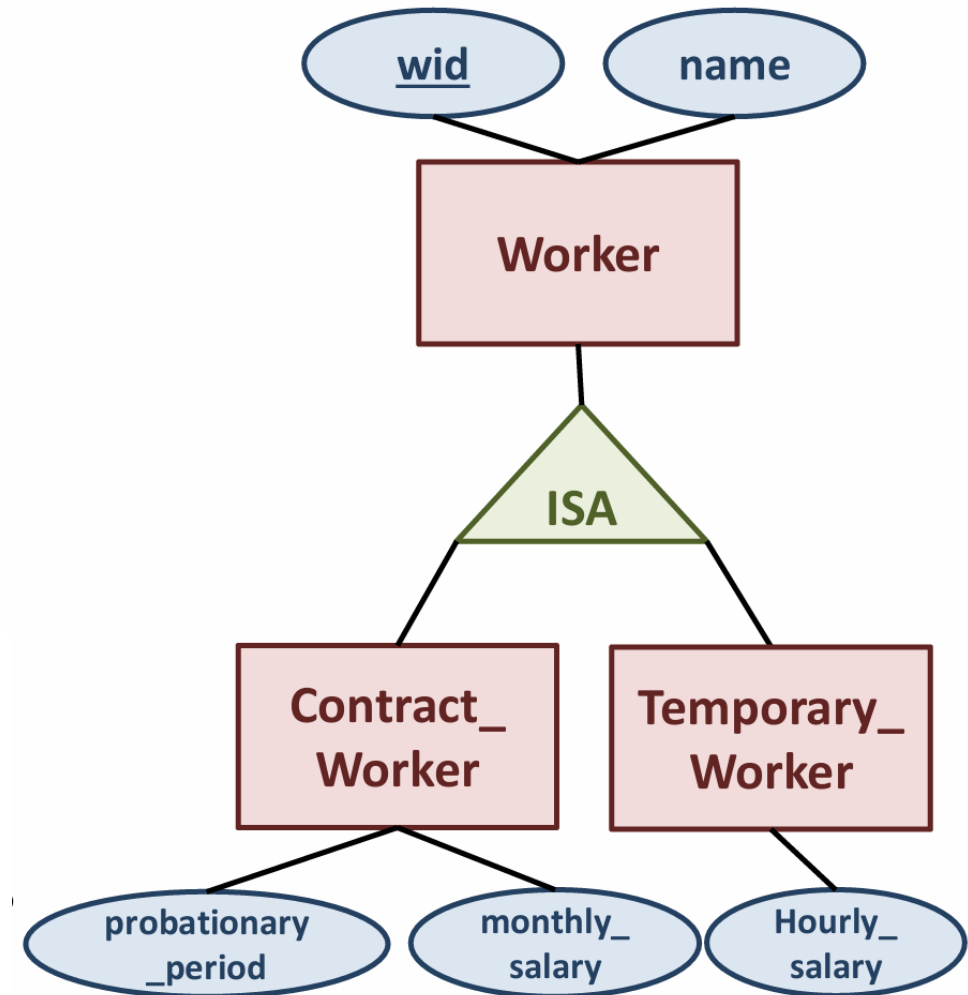
Some Additional Concepts

(EER Features)

Entity Type Hierarchies

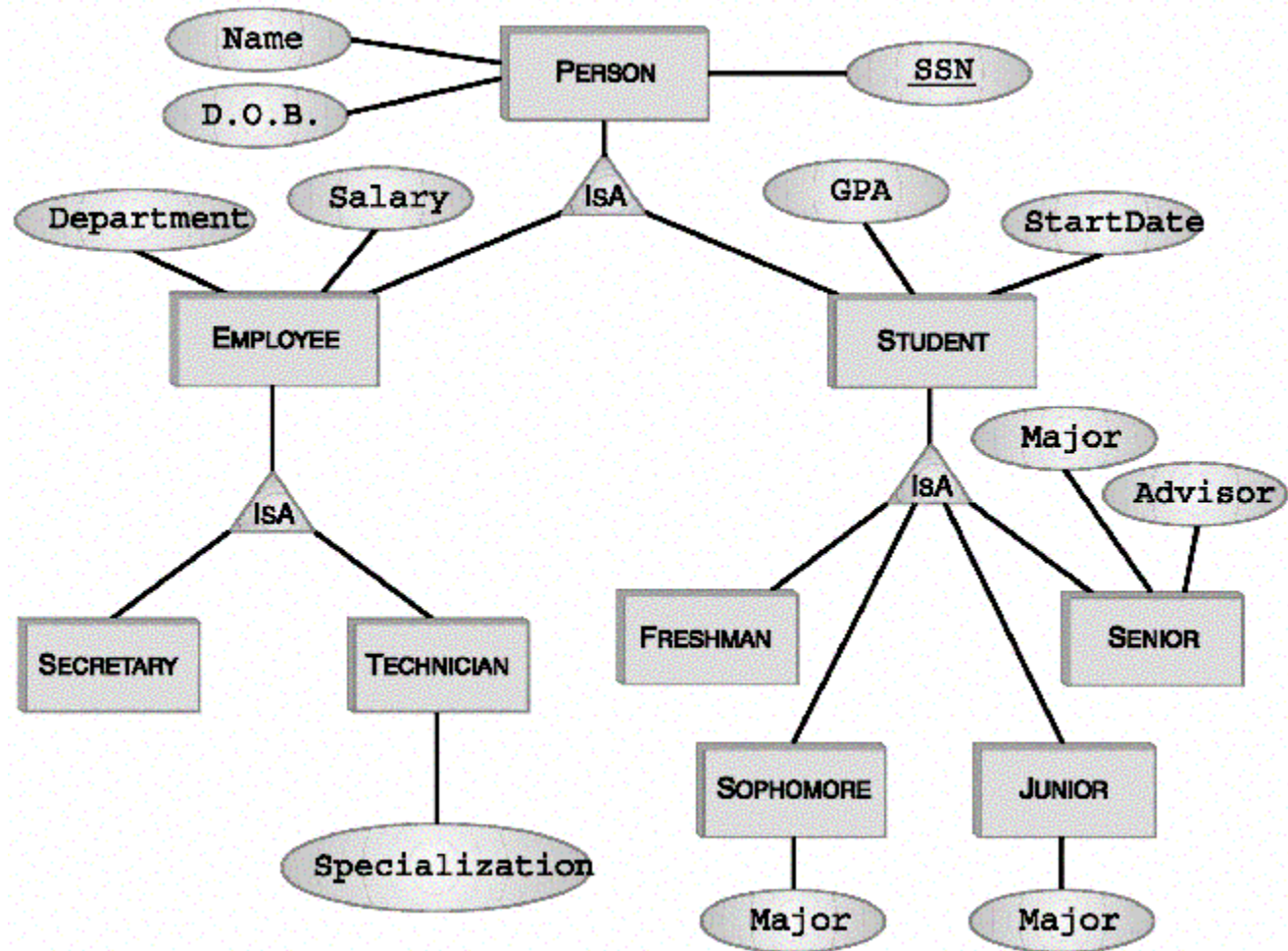
- Classification of an entity as sub-classes
- Considerations as can sub-classes they overlap? Disjoint?

- A contract worker **IS A** worker
attribute set:
{wid, name, probationary_period, monthly salary}
- A temporary worker **IS A** worker
attribute set:
{wid, name, hourly_salary}



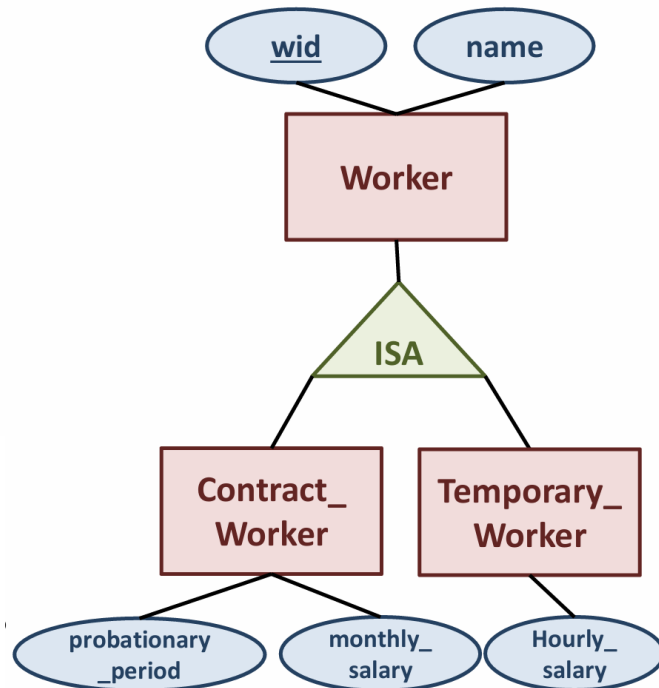
Entity Type Hierarchies

- Inheritance
 - Attributes of supertype apply to subtype
 - creates a hierarchy of **ISA** relationships
- Advantage:
 - more concise and readable E-R diagram
 - It best maps to object oriented approaches (O-O databases)



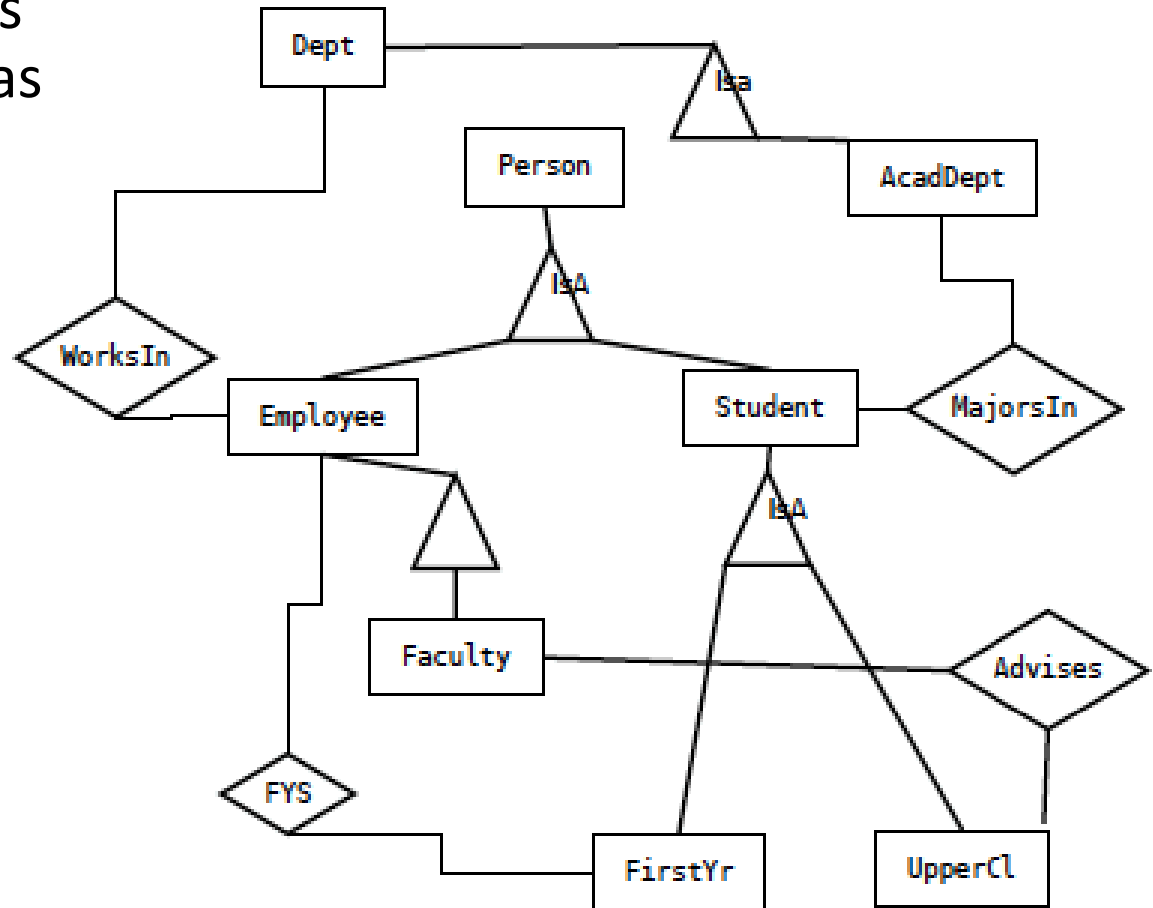
Mapping Specialization & Generalization

- Suggestions:
 - Subclasses Only
 - Superclass and Subclasses
 - Single Relation with Types Attribute (Disjoint)



Entity Type Hierarchies

- Any of the entities or sub-entities can participate in a relationship as appropriate.



Aggregation

- A relationship between a collection of entities and relationships

